Long Yang

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Research Interests

My research focuses on the study of local structure-property relationship in crystalline, non-crystalline, and nanocrystalline materials at the sub-nanometer length scale using x-ray, neutron, and electron diffraction techniques, including pair distribution function (PDF) and atomic electron tomography (AET) methods, by incorporating AI and machine learning algorithms.

Education

PhD in Materials Science and Engineering, Columbia University, New York, USA 2017 - 2021
MSc in Materials Science and Engineering, Columbia University, New York, USA 2015 - 2016
BSc in Information Science and Technology, Fudan University, Shanghai, China 2011 - 2015

Appointments

Professor, School of Materials Science and Engineering, Tongji University, China 2022 - Now
Postdoctoral Scholar, Department of Physics and Astronomy, UCLA, USA 2021 - 2022

Honors & Awards

- Outstanding Mentor at the Tongji University Student Innovation Base, 2023 (Awarded by the School of Innovation and Entrepreneurship at Tongji University)
- Shanghai Leading Talents Program, 2021 (Overseas)
- WangDao Scholar, 2015 (Selected as one of the three distinguished projects at Fudan University)
- Fudan University Scholarship, 2013-2015 (Multiple times)
- National Scholarship, 2012 (Awarded by the Ministry of Education of the People's Republic of China)

Scientific Activities

Community Scientific Software Efforts:

- PDF in the Cloud (PDFitc): A cloud-based total scattering PDF data analysis platform at https://pdfitc.org.
- PDF Database (PDFdata): A open-access PDF database at https://pdfdata.org.
- PDFgui: A local atomic structure modeling software for x-ray and neutron pair distribution function data (Current lead developer).
- ADvanced DIffraction Environment (ADDIE): A cloud-based platform at the Oak Ridge National Laboratory to host neutron diffraction data analysis tools at https://addie.ornl.gov .
- Diffraction Data Pipeline: Time-of-flight neutron diffraction data auto reduction, calibration, and structural fitting of diffraction and PDF datasets collected at the NOMAD beamline of Spallation Neutron Source (SNS).

Committee Members:

- Council Member of the 8th Chinese Crystallographic Society (CCrS), term starts from October 2025.
- Committee member of the local structure and total scattering technique committee of the CCrS, which was founded in October 2023.
- Reviewer for user experiment proposals at International large scientific facilities: Shanghai Synchrotron Radiation Facility (SSRF), China Spallation Neutron Source (CSNS), Japan Proton Accelerator Research Complex (J-PARC).

Journal Reviewers:

- Science Advances, Journal of the American Chemical Society, Acta Materialia, APL Materials, ACS Applied Nano Materials, etc.

TEACHING

- Solid State Physics, 2023 Fall (Graduate)
- Material Science Major Experiments, 2023 Fall, 2024 Fall (Undergraduate)
- Fundamentals of Materials Science, 2024 Fall, 2025 Fall (Undergraduate)
- Material Physics, 2024 Fall, 2025 Fall (Undergraduate)

PATENTS

1. **Long Yang**, Jie Zhou, Hai Qiao, ZL202511799958.6, A database construction method for atomic pair distribution functions.
2. **Long Yang**, Changyuan Li, ZL202610194518.6, An atomic pair distribution function-based method and system for measuring chemical short-range order.
3. Yanzhong Pei, Zhiwei Chen, **Long Yang**, Qingyu Bai, CN202411586052.1, A cesium-silver-iodine insulation material with ultra-low thermal conductivity, and its preparation method and application.

Talks & Conferences

1. *PDF Local Structure Characterization and Low Thermal Conductivity Materials,* The 3rd Youth Academic Conference on Ceramics of the Chinese Crystallographic Society (CCrS) and the Symposium on Crystalline Composite Materials, Kunming, Yunnan, China, April 2026 (Invited Talk)
2. *Recent Development in PDF Methodology: Single-Crystal Diffuse Scattering Technique and Open-Access Database,* The 10th Chinese Crystallographic Society (CCrS) meeting, Guilin, Guangxi, China, October 2025 (Invited Talk, Committee Member)
3. *The Development of Three-Dimensional Pair Distribution Function at SSRF,* Asia-Oceania Conference on Synchrotron Radiation Instrumentation 2025 (AO-SRI 2025, International), Hangzhou, Zhejiang, China, October 2025 (Oral Talk)
4. National Key Research and Development Program of China 'Study on the Structure-Property Relationship of Multiphase Catalysis Based on Neutron Scattering and Spectroscopy' 2025 Midterm Project Self-Evaluation Meeting, ShanghaiTech University, Shanghai, China, June 2025 (Invited Committee Member)
5. *Local Structures of Materials and Atomic Pair Distribution Function,* Sino-French Colloquium Energy Materials for Sustainable Development, Tongji University, Shanghai, China, May 2025 (Oral Talk)
6. *Atomic Local Structures and Pair Distribution Function,* Harbin Institute of Technology, Shenzhen, Guangdong, China, December 2024 (Invited Talk)
7. *PDF Local Structure Characterization and Chemical Short-Range Ordering,* The 13th China Congress on Particle Technology, Suzhou, Jiangsu, China, October 2024 (Invited Talk)
8. *Atomic Local Structures and Low Lattice Thermal Conductivity Materials,* The 16th Chinese Conference on Thermoelectric Materials and Applications, Chongqing, China, October 2024 (Invited Talk)
9. *PDF Local Structure Generative Model Algorithms,* Committee Member Meeting of the Local Structure and Total Scattering Technique Committee of the Chinese Crystallographic Society, Xining, Qinghai, China, July 2024 (Invited Talk, Committee Member)
10. *The Atomic Local Structure Characterization on Materials,* Suzhou Laboratory, Suzhou, Jiangsu, China, June 2024 (Invited Talk)
11. *The Development of Light Sources and Experimental Methods,* 2024 Spring Semester Nanjing University Of Science And Technology International Course on the Introduction to Synchrotron Radiation and Neutron Scattering, Remote, China, May 2024 (Invited Talk)
12. National Key Research and Development Program of China 'Study on the Synchrotron Radiation High-Throughput Characterization of Novel Actinide Alloys' 2024 Mid-term Review Meeting, Shanghai, China, April 2024 (Invited Committee Member)
13. National Key Research and Development Program of China 'Study on the Structure-Property Relationship of Multiphase Catalysis Based on Neutron Scattering and Spectroscopy' 2023 Annual Work Summary Meeting, Dongguan, Guangdong, China, January 2024 (Invited Committee Member)
14. *Unveiling local magnetic correlations: the development of magnetic pair distribution function at CSNS,* The 24th meeting of The International Collaboration of Advanced Neutron Sources (ICANS XXIV, International), Dongguan, Guangdong, China, October 2023 (Invited Talk)
15. *The cloud platform for PDF data analysis of local structure characterization,* The 9th Chinese Crystallographic Society (CCrS) meeting, Fuzhou, Fujian, China, October 2023 (Invited Talk, Committee Member)
16. 2023 China Spallation Neutron Source (CSNS) Multi-Physics Instrument User Meeting, Xining, Qinghai, China, August 2023 (Attendee)
17. *Data Processing for Advanced Light Source Experiments,* IHEP School of Computing 2023, IHEP of CAS, Beijing, China, August 2023 (Invited talk)

18. *The cloud platform for PDF data analysis of local structure characterization,* Chinese Materials Conference 2022-2023 by Chinese Materials Research Society(C-MRS), Shenzhen, Guangdong, China, July 2023 (Outstanding talk award)
19. *The Structure-Property Relationship of Atomic Local Structures and Low Lattice Thermal Conductivities,* The 14th Chinese Conference on Thermoelectric Materials and Applications (CCT2022), Tongxiang, Zhejiang, China, March 2023 (Oral presentation)
20. 2023 China Spallation Neutron Source (CSNS) User Meeting, The Institute of High Energy Physics (IHEP) of the Chinese Academy of Sciences (CAS), Dongguan, Guangdong, China, March 2023 (Attendee)
21. National Program on Key Research and Development 'Basic Research Conditions and Major Scientific Instruments and Equipment Research and Development' Key Projects 2023 Annual Progress Meeting (Scientific Data Section), Beijing, China, March 2023 (Invited committee members)
22. *Atomic Pair Distribution Function Data Processing for Total Scattering Experiments,* 3rd Advanced Light Source Neutron Source Scientific Data and Software Workshop, IHEP of CAS, Beijing, China, March 2023 (Oral presentation)
23. National Key Research and Development Program of China 'Study on the Structure-Property Relationship of Multiphase Catalysis Based on Neutron Scattering and Spectroscopy' Project Initiation and Implementation Plan Demonstration Meeting, Beijing, China, February 2023 (Invited committee members)
24. *The Atomic Local Structure Characterization on Materials,* Seminar at The State Key Lab of High Performance Ceramics and Superfine Microstructure Shanghai, Institute of Ceramics, Chinese Academy of Sciences, Shanghai, China, December 2022 (Invited talk)
25. 2022 Shanghai Soft X-Ray Free Electron Laser Instrument User Meeting, ShanghaiTech University, Shanghai, China, November 2022 (Attendee)
26. *Data Processing for Synchrotron X-ray and Neutron Experiments,* IHEP School of Computing 2022, IHEP of CAS, Beijing, China, August 2022 (Invited talk)
27. *Data Mining and High-Energy Diffraction Data Analysis in the Cloud,* IHEP Computed Center, High Energy Photon Source (HEPS), IHEP of CAS, Beijing, China, February 2022 (Invited talk)
28. *Structure-mining: An Automated Tool to Find Candidate Structures from Neutron and X-ray PDF Data,* PDF-2020 workshop, Remote, UK, China & USA, December 2020 (Invited talk)
29. *Structure-mining: An Automated Tool to Find Candidate Structures from Neutron and X-ray PDF Data,* 10th American Conference on Neutron Scattering (ACNS 2020), Remote, USA, July 2020 (Oral presentation)
30. *Towards Machine Learning on Experimental Nanostructure Data,* Columbia University Data Science Day 2020, New York, USA, March 2020 (Oral presentation, cecorded at https://youtu.be/8xwWrRUkhN4)
31. *Introduction of DiffPy-CMI: What It Can Do Beyond PDFgui,* 2nd US School on Total Scattering Analysis, Oak Ridge, USA, August 2018 (Invited talk)
32. *Role of Local Structural Distortions on Mo and V Nitrides with The Pair Distribution Function Analysis,* Quantum Complex Matter 2018 (QCM 2018), Laboratori Nazionali di Frascati, Rome, Italy, June 2018 (Poster)

Publications

Google Scholar: https://scholar.google.com/citations?user=H2mOKp8AAAAJ

Journal Articles and Books

[1] Jiarui Li, Quan Nie, Te Kang, Ge Qu, Fangzhou Yang, Wang Wan, **Long Yang**, Chao Wang, and Yunhui Huang. Ultrafast joule-heating disproportionation for engineering sub-2 nm Si nanodomains toward stable, high-performance SiO anodes, *J. Am. Chem. Soc.*, 148(25):26017–26026, July 2026. doi: 10.1021/jacs.6c04827.

[2] Ziyue Liu, Qingyu Bai, Zhiwei Chen, Linjie Wu, Te Kang, Changyuan Li, **Long Yang**, Jun Luo, and Yanzhong Pei. Extraordinary heat insulation in $RbAg_4I_5$, *Natl. Sci. Rev.*, 13(13):nwag318, July 2026. doi: 10.1093/nsr/nwag318.

[3] Siyuan Zhang, Jiacong Li, Yuge Cao, Yifeng Zhao, Zhongxing Xu, Zhuoran Lv, **Long Yang**, Chaohong Guan, Zhangliu Tian, Wujie Dong, Haijie Chen, and Fuqiang Huang. Cyclic $P_3O_9^{3-}$ Trimer: A Network Former for Amorphous Superionic Conductors in Sodium Solid-State Batteries, *Angew. Chem. Int. Ed.*, page e4295699, July 2026. doi: 10.1002/anie.4295699.

[4] Yuhao Jin, Hao Fang, Haoyu Pan, Zhenyi Zhang, Chaofeng Liu, **Long Yang***, Zheng Zhou*, and Haixiang Han*. Single-Atom Migration into a Chiral Multilayer Nanocluster Architecture, *J. Am. Chem. Soc.*, June 2026. doi: 10.1021/jacs.6c03831.

[5] Shujing Yang, Shanshan Hu, Ziyan Li, Changyuan Li, **Long Yang**, Zhiwei Chen, Juan Chen, Jun Luo, Wen Li, and Yanzhong Pei. Plastic processing induced multiscale microstructures for simultaneous thermoelectric and mechanical enhancements in PbTe, *Sci. China Mater.*, June 2026. doi: 10.1007/s40843-026-4345-7.

[6] Linjie Wu#, Pengfei Nan#, **Long Yang#**, Zhiwei Chen, Changyuan Li, Qingyu Bai, Hongzheng Wang, Sui Ting Tai, Chen Wang, Binghui Ge, Yue Chen, Jun Luo, and Yanzhong Pei. Flapping vibrations contributed to heat insulation in Cs_2ZnI_4, *Natl. Sci. Rev.*, page nwag256, May 2026. doi: 10.1093/nsr/nwag256.

[7] Changyuan Li, Di Zhang, Xiaoyu Zhang, Hongyao Zhang, Jiang Cheng, Zhiwei Chen, He Lin, Xiang Meng*, Wen Li*, and **Long Yang***. Nanoscale local symmetry optimization induced low lattice thermal conductivity in I-V-VI_2 compounds, *Adv. Funct. Mater.*, 36(41):e74755, March 2026. doi: 10.1002/adfm.74755.

[8] Yiyuan Tao, Xingyu Zheng, Shi Huang, Ershuai Liu, Yangjian Lin, Qike Jiang, **Long Yang**, Walter S. Drisdell, Tianding Xu, and Yao Yang. Multi-modal characterization of nitrate reduction nano-catalysts with periodic strain distribution, *Nat. Commun.*, 17(1):3778, March 2026. doi: 10.1038/s41467-026-70447-7.

[9] Xinjue Zhang, Te Kang, Heng Liu, Changyuan Li, Huanhuan Niu, Chenlong Gao, Menghao Yang, **Long Yang***, Guozhong Cao*, and Chaofeng Liu*. Visualizing ion transport pathways to unveil the origins of fast ion conduction in copper-ion solid electrolyte for all-solid-state batteries, *Small Methods*, page e02312, February 2026. doi: 10.1002/smtd.202502312.

[10] Te Kang, Changyuan Li, Xinjue Zhang, Yuiga Nakamura, Jo-chi Tseng, Taishun Manjo, Chaofeng Liu, and **Long Yang***. Local symmetry breaking induced superionic conductivity in argyrodites, *J. Am. Chem. Soc.*, 148(6):6158–6166 (Selected as Journal Cover), January 2026. doi: 10.1021/jacs.5c17193.

[11] Jidong Ma, Heng Liu, Te Kang, Changyuan Li, Huanhuan Niu, **Long Yang**, Chaofeng Liu, and Guozhong Cao. Controlling ligand field of Li_3VO_4 to enhance the electrochemical performance for lithium-ion batteries, *Sci. China Mater.*, 69(2):838–847, January 2026. doi: 10.1007/s40843-025-3712-2.

[12] Jiajia Lu, Sheng Zhou, **Long Yang**, Yue Xi, Yuwei Li, Kun Zhang, Nan Lu, and Jianjun Yang. Reticulocalbin-2–Programmed Macrophage Exosomes Restore Immuno-Osteogenic Coupling to Promote Calvarial Bone Regeneration, *Adv. Funct. Mater.*, page e76804, 2026. doi: 10.1002/adfm.76804.

[13] Yuwei Zhao, Wujie Dong, Shiyu Zhang, Xuyang Li, Kehong Wang, Minghui Yu, Zhuoran Lv, Song Liang, Meibing Chen, Yan Zhang, Sixiao Guo, **Long Yang**, and Fuqiang Huang. Partial Atomic Disordered $Ti_2Nb_{10}O_{29-x}$ Induced by Joule Thermal Shock for Superior Lithium-Ion Storage, *Adv. Mater.*, 38(39):e73607, 2026. doi: 10.1002/adma.73607.

[14] Heng Liu, Menghao Yang, Quan Zong, Min-Hsin Yeh, Chun-Chi Chang, **Long Yang**, Wei-Hsiang Huang, Chaofeng Liu, and Guozhong Cao. Suppressed proton insertion enhances zinc-ion storage kinetics and stability in hydrated vanadate, *ACS Energy Lett.*, 11(1):644–653, December 2025. doi: 10.1021/acsenergylett.5c03338.

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[17] Jing Li, Hongyao Zhang, Changyuan Li, Zhijun Wang, Tongsuo Lu, **Long Yang***, and He Lin*. Application and development of three-dimensional differential pair distribution function method in Shanghai Synchrotron Radiation Facility, *Nucl. Tech.*, 48(10):100101, October 2025. doi: 10.3724/j.0253-3219.2025.hjs.48.240481.

[18] Shanshan Hu, Chen Huang, Changyuan Li, **Long Yang**, Zhiwei Chen, Baisheng Sa, Wen Li, and Yanzhong Pei. Thermal stability of FeSi as barrier layer in high-performance $Mg_2Si_{0.3}Sn_{0.7}$ thermoelectric device, *J. Materiomics*, 11(5):101044, September 2025. doi: 10.1016/j.jmat.2025.101044.

[19] Haitao Wang, Te Kang, **Long Yang***, and Yanzhong Pei*. Deciphering the functions of rubidium in structural stability and ionic conductivity of KAg_4I_5, *J. Mater. Chem. A*, 13(38):32179–32186, September 2025. doi: 10.1039/D5TA03873A.

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[23] Heng Liu#, **Long Yang#**, Ting Shen, Changyuan Li, Te Kang, Huanhuan Niu, Wei-Hsiang Huang, Chun-Chi Chang, Menghao Yang, Guozhong Cao, and Chaofeng Liu. Distorting Local Structures to Modulate Ligand Fields in Vanadium Oxide for High-Performance Aqueous Zinc-Ion Batteries, *ACS Nano*, 19(9):9132–9143, March 2025. doi: 10.1021/acsnano.4c18250.

[24] Di Zhang, Min Liu, Tao Jin, **Long Yang***, Wen Li*, and Yanzhong Pei*. $(AgSbTe_2)_{1-x}(Bi_2Te_3)_x$-based thermoelectric device for low-grade heat recovery, *Mater. Today Phys.*, 52:101692, March 2025. doi: 10.1016/j.mtphys.2025.101692.

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[28] **Long Yang***, Juping Xu*, and Wen Yin. Unveiling Short-Range Magnetic Correlations: The Development of Magnetic Pair Distribution Function Method at CSNS, *Nucl. Instrum. Methods Phys. Res. Sect. Accel. Spectrometers Detect. Assoc. Equip.*, 1069:169967, December 2024. doi: 10.1016/j.nima.2024.169967.

[29] Yixin Hu, Xinyi Shen, Zhiwei Chen, Min Liu, Xinyue Zhang, **Long Yang**, Jun Luo, Wen Li, and Yanzhong Pei. Superior bendability enabled by inherent in-plane elasticity in Bi_2Te_3 thermoelectrics, *Mater. Today Phys.*, 48:101570, November 2024. doi: 10.1016/j.mtphys.2024.101570.

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[31] Heng Liu, Huanhuan Niu, Wei-Hsiang Huang, Ting Shen, Changyuan Li, Chun-Chi Chang, Menghao Yang, Chenlong Gao, **Long Yang**, Quan Zong, Yanzhong Pei, Guozhong Cao, and Chaofeng Liu. Unveiling the Local Structure and the Ligand Field of Organic Cation Preintercalated Vanadate Cathode for Aqueous Zinc-Ion Batteries, *ACS Energy Lett.*, 9(11):5492–5501, October 2024. doi: 10.1021/acsenergylett.4c02709.

[32] Huanhuan Niu, Heng Liu, **Long Yang**, Te Kang, Ting Shen, Bingqi Jiang, Wei-Hsiang Huang, Chun-Chi Chang, Yanzhong Pei, Guozhong Cao, and Chaofeng Liu. Impacts of distorted local chemical coordination on electrochemical performance in hydrated vanadium pentoxide, *Nat. Commun.*, 15(1):9421, October 2024. doi: 10.1038/s41467-024-53785-2.

[33] Xiaoyu Zhang, Changyuan Li, Yuting Meng, **Long Yang***, and Wen Li*. Intrinsically Atomic-Disordered $LuCuTe_2$ as a Promising Thermoelectric Material, *ACS Appl. Energy Mater.*, 7(14):5728–5735, July 2024. doi: 10.1021/acsaem.4c00763.

[34] Julien Lombardi, **Long Yang**, Nasim Farahmand, Anthony Ruffino, Ali Younes, Jonathan E. Spanier, Simon J. L. Billinge, and Stephen O'Brien. Structure and phase transitions in niobium and tantalum derived nanoscale transition metal perovskites, $Ba(Ti,M^V)O_3$, M=Nb,Ta, *J. Chem. Phys.*, 160(13):134702, April 2024. doi: 10.1063/5.0192488.

[35] Tao Jin#, **Long Yang#**, Xinyue Zhang, Wen Li, and Yanzhong Pei. Close-packed layer spacing as a practical guideline for structure symmetry manipulation of IV-VI/I-V-VI_2 thermoelectrics, *InfoMat*, 6(2):e12502, February 2024. doi: 10.1002/inf2.12502.

[36] Saman Moniri, Yao Yang, Jun Ding, Yakun Yuan, Jihan Zhou, **Long Yang**, Fan Zhu, Yuxuan Liao, Yonggang Yao, Liangbing Hu, Peter Ercius, and Jianwei Miao. Three-dimensional atomic structure and local chemical order of medium- and high-entropy nanoalloys, *Nature*, 624(7992):564–569, December 2023. doi: 10.1038/s41586-023-06785-z.

[37] Yuchen Shang, Mingguang Yao, Zhaodong Liu, Rong Fu, Longbiao Yan, **Long Yang**, Zhongyin Zhang, Jiajun Dong, Chunguang Zhai, Xuyuan Hou, Liting Fei, GuanJie Zhang, Jianfeng Ji, Jie Zhu, He Lin, Bertil Sundqvist, and Bingbing Liu. Enhancement of short/medium-range order and thermal conductivity in ultrahard sp^3 amorphous carbon by C_{70} precursor, *Nat. Commun.*, 14(1):7860, November 2023. doi: 10.1038/s41467-023-42195-5.

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[39] Simon J. L. Billinge, Sandra H. Skjærvø, Maxwell W. Terban, Songsheng Tao, **Long Yang**, Yevgeny Rakita, and Benjamin A. Frandsen. Local structure determination using total scattering data. In *Comprehensive Inorganic Chemistry III*, pages 222–247. Elsevier, Oxford, third edition edition, January 2023.

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[41] Zachary Thatcher, Chia-Hao Liu, **Long Yang**, Brennan C. McBride, Gia Thinh Tran, Allison Wustrow, Martin A. Karlsen, James R. Neilson, Dorthe B. Ravnsbæk, and Simon J. L. Billinge. nmfMapping: A

cloud-based web application for non-negative matrix factorization of powder diffraction and pair distribution function datasets, *Acta Crystallogr. Sect. Found. Adv.*, 78(3):242–248, May 2022. doi: 10.1107/S2053273322002522.

[42] Muhammad Boota, Tanveer Hussain, **Long Yang**, Matthieu Bécuwe, William Porzio, Luisa Barba, and Rajeev Ahuja. Mechanistic Understanding of the Interactions and Pseudocapacitance of Multi-Electron Redox Organic Molecules Sandwiched between MXene Layers, *Adv. Electron. Mater.*, 7(4):2001202, March 2021. doi: 10.1002/aelm.202001202.

[43] **Long Yang**, Elizabeth A. Culbertson, Nancy K. Thomas, Hung T. Vuong, Emil T. S. Kjær, Kirsten M. Ø Jensen, Matthew G. Tucker, and Simon J. L. Billinge. A cloud platform for atomic pair distribution function analysis: PDFitc, *Acta Crystallogr. Sect. Found. Adv.*, 77(1):2–6, January 2021. doi: 10.1107/S2053273320013066.

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[46] **Long Yang**, Pavol Juhás, Maxwell W. Terban, Matthew G. Tucker, and Simon J. L. Billinge. Structure-mining: Screening structure models by automated fitting to the atomic pair distribution function over large numbers of models, *Acta Crystallogr. Sect. Found. Adv.*, 76(3):395–409, May 2020. doi: 10.1107/S2053273320002028.

[47] Xu Xiao, Hao Wang, Weizhai Bao, Patrick Urbankowski, **Long Yang**, Yao Yang, Kathleen Maleski, Linfan Cui, Simon J. L. Billinge, Guoxiu Wang, and Yury Gogotsi. Two-Dimensional Arrays of Transition Metal Nitride Nanocrystals, *Adv. Mater.*, 31(33):1902393, June 2019. doi: 10.1002/adma.201902393.

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